

# Validation of the Ideal Gas Law Using Molecular Dynamics Simulations

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**Abstract.** This project investigates the emergence of macroscopic thermodynamic properties from microscopic particle dynamics through a molecular dynamics simulation of an ideal gas. By modelling elastic collisions between particles and container walls, pressure and temperature are computed directly from particle motion and compared with the predictions of the Ideal Gas Law. The simulation is also used to study the influence of numerical parameters and initial conditions on accuracy, as well as the emergence of the Maxwell-Boltzmann velocity distribution at equilibrium.

## 1 Introduction

One of the central goals of statistical mechanics is to explain macroscopic thermodynamic laws from microscopic particle behaviour. The Ideal Gas Law,

$$PV = Nk_B T, \quad (1)$$

relates pressure, volume, temperature, and particle number through a simple equation of state. Despite its simplicity, this law emerges from the collective dynamics of a large number of particles.

The cornerstone of this simulation is the Kinetic Theory of gases, which relates macroscopic and microscopic properties of a closed system. The theory states that a gas can be treated as a collection of many small particles, and that properties such as temperature and pressure are macroscopic manifestations of particle velocities and interactions within the volume. To demonstrate the relations established by kinetic theory, we verify the Ideal Gas Law, a fundamental relation between pressure, volume, and temperature used extensively in chemistry and physics.

## 2 Simulation Implementation

The simulation was implemented in Python using:

- **NumPy:** used for array operations and mathematical calculations.
- **Matplotlib:** used to visualise the data produced by the simulation.
- **Itertools:** used to assign particle IDs and to generate all particle pairs.
- **Ipycanvas:** used to animate the particles.

The simulation uses the Euler integration method [1] to compute the position, momenta, and collisions of each particle, and then uses these results to calculate the pressure and temperature of the system.

Given the assumptions of the ideal gas model, the simulation rests on the following key premises:

- The gas consists of identical monoatomic particles.
- Particle interactions occur only during collisions.

- All collisions are perfectly elastic.
- External forces are neglected.
- The system is closed and energy is conserved.

Under these assumptions, pressure arises from momentum transfer during particle wall collisions, while temperature is related to the average kinetic energy of the particles.

From basic dynamics, the infinitesimal displacement over a small time step is  $d\vec{x}(t) = \vec{v}_0 dt$ .

Collision detection checks whether the distance between two particles is less than the sum of their radii. When a collision is detected, the elastic collision equations are applied to update the particle velocities. The new velocity of particle 1 after the collision is [3]:

$$\vec{v}'_1 = \vec{v}_1 - \frac{(\vec{v}_1 - \vec{v}_2) \cdot (\vec{r}_1 - \vec{r}_2)}{|\vec{r}_1 - \vec{r}_2|^2} (\vec{r}_1 - \vec{r}_2). \quad (2)$$

The simulation loop consists of the following steps:

1. Particle positions are updated.
2. Particle particle collisions are detected.
3. Elastic collision equations are applied.
4. Particle wall collisions are resolved.
5. Momentum transfer to the walls is accumulated.

## 3 Pressure and Temperature

The equipartition theorem [2] states that each degree of freedom contributes  $\frac{1}{2}k_B T$  to the average kinetic energy. For a three-dimensional monoatomic gas, each particle has three translational degrees of freedom. Combining this with the definition of kinetic energy gives:

$$T = \frac{m\langle v^2 \rangle}{3k_B}. \quad (3)$$

Pressure follows from  $P = F/A$ . The force exerted on the container wall is obtained via Newton's second law:

$$|\vec{F}| = \left| \frac{d\vec{p}}{dt} \right| = \frac{1}{\Delta t} \sum |\Delta \vec{p}|, \quad P = \frac{F}{A} = \frac{1}{A \Delta t} \sum |\Delta \vec{p}|.$$

By conservation of momentum, the momentum transferred to a wall per particle collision is  $p_{\text{wall}} = 2m_p v_p$ .

For a cubic container of side  $l$  with total surface area  $A = 6l^2$ , the pressure becomes:

$$P = \frac{1}{3l^2 \Delta t} \sum m_p v_p.$$

Combining both results transforms equation (1) into:

$$PV = Nk_B T = \frac{Nm\langle v^2 \rangle}{3}, \quad (4)$$

which allows a direct comparison with the theoretical prediction of the Ideal Gas Law.

## 4 Sensitivity Analysis

The simulation relies on several assumptions and numerical parameters, so it is important to understand how these affect the results. Of particular importance are the time step, which determines the resolution of the integration, and the particle number and radius, since particles that are too large or a gas that is too dense will deviate from the ideal-gas assumption.

### 4.1 Time Step

The time step determines both the accuracy and the stability of the simulation, as well as the runtime, since smaller time steps require more iterations to reach the same physical time. The fixed parameters used in this test are given in Table 1.

Table 1: Fixed simulation parameters for the sensitivity analysis.

| Parameter             | Value |
|-----------------------|-------|
| Number of particles   | 300   |
| Container side length | 1.0   |
| Particle radius       | 0.001 |
| Particle mass         | 0.001 |
| Mean velocity         | 500   |

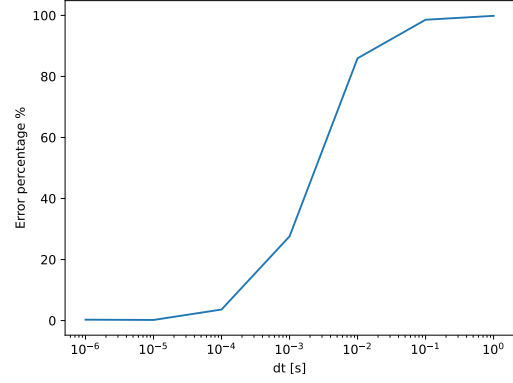
The time steps tested were  $10^i$ ,  $i \in [-6, 0]$ . Figure 1a shows that the error remains small for time steps below  $10^{-4}$ ; this value is used throughout the remainder of the study.

### 4.2 Particle Radius

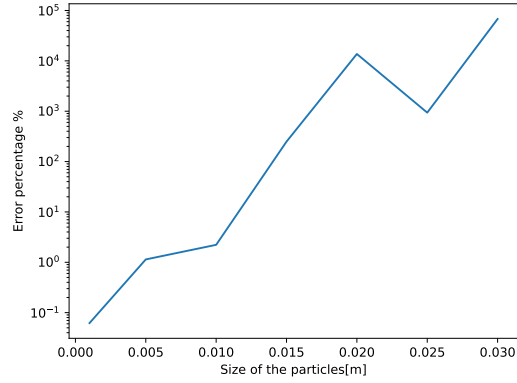
Keeping all other parameters fixed and varying the particle radius (Figure 1b) shows that any radius above  $R = 0.001$  produces unacceptably large errors.

### 4.3 Mean Initial Velocity

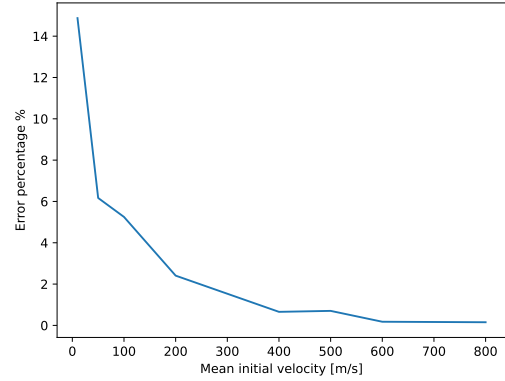
Keeping all other parameters fixed and varying the mean initial velocity (Figure 1c) shows that for velocities above  $600 \text{ m s}^{-1}$  the returns diminish rapidly, so larger values are not warranted.



(a) Time-step sensitivity.



(b) Particle-radius sensitivity.



(c) Mean-velocity sensitivity.

Figure 1: Sensitivity analysis of simulation parameters.

## 5 Optimised Simulation

The following configuration was found to provide reliable results:

- Time step:  $10^{-5} \text{ s}$
- Number of particles: 300
- Particle radius: 0.001 m
- Mean initial velocity:  $600 \text{ m s}^{-1}$

These parameters produced stable dynamics and accurate thermodynamic measurements.

## 6 Results

Using the optimised configuration, the measured pressure and temperature satisfied the Ideal Gas Law with a final relative error below 5%.

This result demonstrates that macroscopic thermodynamic behaviour emerges naturally from microscopic particle interactions, thereby validating the assumptions of kinetic theory within the limits of the simulation. Particle animations further confirmed the expected random motion and uniform spatial distribution characteristic of an equilibrium gas.

## References

- [1] John C. Butcher. *Numerical Methods for Ordinary Differential Equations*. Chichester: John Wiley & Sons, 2003, p. 45. ISBN: 978-0-471-96758-3.
- [2] Charles Kittel and Herbert Kroemer. *Thermal Physics*. W. H. Freeman, 1980.
- [3] Dennis C. Rapaport. *The Art of Molecular Dynamics Simulation*. 2nd ed. Cambridge University Press, 2004.